

	DATASHEET		N° I-FD-5270.00-52313-341-B14-101 FD-REPLAN-5283-33-001						
	CLIENT: REPLAN - REFINARIA DE PAULÍNIA		SHEET: 1 of 6						
	JOB: HDT IV DE MÉDIOS DA REPLAN								
	ÁREA: U-5283-UNIDADE DE HIDROTATAMENTO DE DIESEL IV								
CONSÓRCIO TOYO SETAL HDT PAULÍNIA	TITLE: SPECIAL PURPOSE GEAR VV-C-5283001		INTERNO						
			SRGE/SI-III/REPLAN						
	RM N°:	I-RM-5270.00-52313-321-TS8-001-0	TEC. RESPONSIBLE:	Maurizio Zipoli					
	OC N°:	BA130300-AXGB001							
	VENDOR N°:	YOH000090231	CREA N°:01	N/A					
	ELETRONIC FILE:	I-FD-5270.00-52313-341-B14-101=0.pdf							
INDEX OF REVISION									
REV.	DESCRIPTION AND/OR THE REVISED SHEETS								
0	Issue for review								
A	Issue for review								
B	Issue for review								
C	Issue for review								
D	Issue for review								
E	Certified								
Consórcio TOYO SETAL HDT Paulínia A APPROVED WITH NO COMMENTS Update revision as CERTIFIED. Start the fabrication. B APPROVED WITH COMMENTS Resubmit for approval. Do not start the fabrication. C APPROVED WITH COMMENTS Revise document to CERTIFIED. Start the fabrication. D DISAPPROVED Resubmit for approval. Do not start the fabrication. X E CERTIFIED ACCEPTED F FOR INFORMATION G CANCELLED Ass: M.BATISTA The comments on this document do not excuse the supplier from the responsibilities from the project, fabrication and equipment performance. MARCELO CASTRO BATISTA:040401346 56 Assinado de forma digital por MARCELO CASTRO BATISTA:04040134656 Dados: 2023.12.15 15:19:49 -03'00'									
	REV. 0	REV. A	REV. B	REV. C	REV. D	REV. E	REV. F	REV. G	REV. H
DATE	21/03/2023	11/05/23	12/07/23	12/09/23	03/11/23	11/12/23			
DESIGN	BI4	BI4	BI4	BI4	BI4	BI4			
EXECUTION	CD	CD	CD	CD	CD	ES			
CHECK	TC	OO	OO	OO	OO	JM			
APPROVAL	OO	EH	MZ	MZ	MZ	MZ			
INFORMATION IN THIS DOCUMENT IS PROPERTY OF PETROBRAS, BEING PROHIBITED OUTSIDE OF THEIR PURPOSE									
FORM OWNED TO PETROBRAS N-0381 REV. L									

		NF1819E API DATA SHEET				YOH000090231	
GEARBOX DATA SHEET							
		TITLE: NF1819E API DATA SHEET			DOCUMENT CODE YOH000090231		REVISION 6
REVISION DESCRIPTION: Revised at customer request			REVISION DATE 11-Dec-23		APPROVED Electronically Stored		SECURITY CODE N
					CHECKED Electronically Stored		
					EXECUTED SBARCEA, Elisabeta		
	SCALE N/A	REPLACES/DERIVED FROM N/A		1 st EXECUTION 21-Mar-23	ORIGINAL JOB 01864H1	SIZE 4	LANGUAGE A
© Unpublished Work of Lufkin Gears France, S.A.S, an affiliate of Baker Hughes ("BH"). This document contains proprietary and confidential information not to be distributed, transmitted, reproduced, altered, published, released, or used except as authorized by BH, and is subject to return on demand.							SHEET 2 of 6

SPECIAL PURPOSE GEAR UNITS **API 613 FIFTH EDITION** **DATA SHEET** **SI UNITS**

Job No. 01864H1 Item No. VV-C-5283001
P.O. No. 440591464 Date 20-janv-23
Requisition No. _____
Inquiry No. VV-C-5283001
Doc No. YOH000090231 Rev 6 Date 11/12/2023 By LFK

1 Applicable To: ☐ Proposal ☐ Purchase ☒ As Built
2 For REPLAN
3 Site REPLAN HDT IV PAULINA-SP-BRASIL
4 Unit U-5283
5 Service RECYCLE GAS COMPRESSOR C-5283001
6 No. Required 1

☐ US Standards (1.7) ☒ ISO Standards (1.7)
Manufacturer LUFKIN
Model No. NF1819E
Serial No. YH00558
Driver Type Variable Speed Motor
Driven Equipment Centrifugal Compressor

NOTE: Numbers Within () Refer To Applicable API Standard 613 Paragraphs

☒ Information To Be Completed By Purchaser

☐ Information To Be Completed By Manufacturer

UNITS OF MEASUREMENT (1.8)

☐ US Customary Units
☒ SI Units
RATING REQUIREMENTS
Driven Equip. kw (2.1.4): Norm. 1669 Max 2120
Driver kw: Rated 2474 Max 2474
Normal Transmitted Power (2.2.2) 1562 kw
Gear Unit Rated Power (2.2.1) 2532 kw
Torque @ Max Cont Speec 24761 N m
Max Torque (2.2.1) 67473 N m @ 1953 RPM
☐ Reducer ☒ Increaser

Rated Speed, RPM (2.1.5) :
Input 1953 ☒ Specified ☐ Nominal
Output 12224 ☐ Specified ☒ Nominal
Allow Var In Gear Ratio (2.1.5) (+)(-) 0,1 %
Max Continuous Speed (1.5) 12835 RPM
Trip Speed (1.5)(2.1.7) 14118 RPM
Gear Service Factor (2.2.3.1) 1,6 (Min)
Hardness (2.2.3.2) Pinion 58 RC Gear 58 RC
Shaft Assembly Designation (2.1.19) R-L
HS Shaft Rot Fac'g Cpl'g (2.1.20.2) ☐ CW ☒ CCW
LS Shaft Rot Fac'g Cpl'g (2.1.20.2) ☐ CW ☒ CCW
External Loads (2.1.16)
Other Operating Conditions (2.2.2) (2.6.1.3)

INSTALLATION DATA (2.1.14)

☐ Indoor ☐ Heated ☒ Under Roof
☒ Outdoor ☒ Unheated ☐ Partial Sides
☒ Grade ☐ Mezzanine ☐
☐ Winterization Req'd ☒ Tropicalization Required
Elec. Area (2.1.10) Class Grp IIC Div
Elec'l Area (2.1.10) Class Zone 2 Temp T1
Max Allow SPL (2.1.6) 85 dBA @ 1 m
Elevation 595 m Barometer 95,12 kPa abs
Range Of Ambient Temperatures :
Dry Bulb Wet Bulb
Normal 22 °C °C
Maximum 38 °C °C
Minimum 1 °C °C
Unusual Conditions ☒ Dust ☒ Fumes
☒ Relative humidity 20% (min), 73% (Nor), 100%(max)

BASIC GEAR DATA

☒ Single Stage ☐ Single Helical
☐ Double Stage ☒ Double Helical
☐ Epicyclic
Mechanical Rating (1.5.16) kw 4051 @ 12224 RPM
Gear Service Factor (2.2.3.1) 1,63 (Actual)
Full Load Gear Unit Power Loss 30 kw
Gear Unit Mechanical Efficiency 98,8 %
Rating Speed, RPM Pinion 12224 Gear 1953
Hardness used for
Rating, (HB or Rc) Pinion 58 Gear 58
Tooth Pitting Index, "K" mpa (2.2.4.1)(2.2.4.2) :
Allowable 1,896 Actual 1,853
Material Index Number, mpa (Fig 3, Table 4) 3,034
Bending stress number, "St" mpa (2.2.5.1)(2.2.5.2)
Pinion Allowable 265,45 Actual 259,8
Gear: Allowable 265,45 Actual 232,54
Pitch Line Velocity 80,62 m/sec
Anticipated SPL (2.1.6) 82 dBA @ 1 m
WR² Referred To LS Shaft 62 kg m²
Breakaway Torque 288 Nm @LS (friction coeff = 0,3)
Pinion teeth Hardness Range 58 - 62
Pinion teeth hardening method CASE (GAS CARBURIZED)
☐ HS Shaft Separate Hardness Range
Gear Teeth (rim) Hardness Range 58 - 62
Gear teeth hardening method: CASE (GAS CARBURIZED)
Gear Hub:
☒ Forged Cylinder ☐ Forged & Coped ☐ Fabricated
Gear To Shaft fit method (2.5.3.2)
☐ Integral ☐ Keyed interference ☒ Keyless Inteference
Rim Attachment (2.5.3.2) INTEGRAL W/HUB
LS Shaft Hardness Range 285 - 331
Journal static Weight loads (2.6.6.3)
Pinion 16 / 24 kg Gear 370 / 555 kg
Total Gear Unit Assembled Weight 2896 kg

NOTES:

SPECIAL PURPOSE GEAR UNITS **API 613 FIFTH EDITION** **DATA SHEET** **SI UNITS**

Job No. 01864H1 Item No. VV-C-5283001
P.O. No. 440591464 Date 20-janv-23
Doc No. YOH000090231 Rev 6 Date 11/12/2023 By LFK

GEAR DATA		COUPLINGS AND GUARDS	
Pinion	Gear	Pinion Shaft	Gear Shaft
Number Of Teeth 27	169	Coupling Furnished By (3.2.1)	BHNP BHNP
Gear Ratio 6,2593		Mount Coupling Halves (3.2.2)	
Tangential Load, "Wt", N (2.2.4.2) 31391		Cplg. Guard Adapter By (3.2.3)	LFK LFK
AGMA Geometry Factor "J": 0,478	0,534	Cplg. Guard Furnished By (3.2.3)	BHNP BHNP
Pitch Diameter, mm 125,96	788,44	Coupling vendor (3.2.3)	KOP FLEX KOP FLEX
Outside Diameter, mm 133,96	796,44	Vendor's model number (3.2.3)	B11142225B B11142225A1
Root Diameter, mm 113,18	775,67	Coupling weight on shaft, kg	7,1 53,1
Center Groove Diameter, mm 110	772	CG Inboard/outboard of shaft end	11,3 90,2
Normal Pressure Angle, Degrees 20		Hub Drill Template Provided	○ ○
Normal Diametral Pitch 6,35	4,00	SHAFT END DETAIL (2.5.4.2)	
Helix Angle, Degrees 30,9749		Shaft end detail specified by: ☒ Purchaser ☐ Gear Vendor	
Center Distance 457,20		(Integral Unless Otherwise Specified)	
Backlash, mm (min. / max.) 0,305 / 0,559		Shaft end detail if 'Otherwise' specified: Pinion Gear	
Net Face Width, "Fw", mm 156,01		Tapered/Keyless ☒ ☐ ☐ ☐	
Pinion L/D 1,61		Tapered / 1-Key ☐ ☐ ☐ ☐	
Face Overlap Ratio 6,389		Tapered / 2-Keys ☐ ☐ ☐ ☐	
Transverse Contact Ratio 1,401		Cylindrical / 1-Key ☐ ☐ ☐ ☐	
AGMA 6011 Service Factor 2,99		Cylindrical / 2-Keys ☐ ☐ ☐ ☐	
Ratings based on ANSI/AGMA 6011 with SF=1.0		Other ☐ Integral Flange ☐ ☒	
Durability Power 8888		Shaft Diameter, mm 60 190	
Strength Power 7583		(If integral flange use diameter immediately adjacent to flange)	
Tooth Surface Finish, μ m RA 0,61	0,61	RADIAL BEARINGS	
Tooth Generation Process HOBBING	HOBBING	Pinion Gear	
Tooth Finishing Process GRINDING	GRINDING	Type HYD. JOURNAL HYD. JOURNAL	
Lead Modification (2.2.4.6) ☐ Not Req'd ☒ Req'd		Diameter, mm 80,00 190,00	
Calculated Total Lead Mismatch, mm (2.2.4.5) <= 0,018		Length, mm 80,00 95,00	
SCUFFING DATA (2.2.6)		Journal Velocity, m/sec 51,2 19,4	
Scuffing Data per ANSI/AGMA 925-A03		Loading, kPa 2664,4 944,7	
Scuffing Risk (2.2.6) LOW Calculation Method AGMA 925		Clearance (min-max), mm 0,178 0,216 0,191 0,249	
Composite surface roughness σ_x 0,50 μ m		Span, mm 350 340	
Specific film thickness, EHL λ_{min} 2,51		Power loss each brg., kw 7,63 2,92	
Tooth temperature, θ_M 70 °C		Oil flow each brg., l/min 19,4 10,15	
Maximum contact temperature, $\theta_{B max}$ 88 °C		THRUST BEARING(S)	
MATERIALS see Note 2 in page 4		Location LSP SHAFT	
Gear Casing S275 J0 Oil Seals Aluminium Marine Grade		Manufacturer KINGSBURY	
Pinion (s) 18CrNiMo7-6		Type JJ	
Gear Rim(s) 18CrNiMo7-6		Size JJ7	
HS Shaft INTEGRAL LS Shaft 42/50CrMo4		Area, mm ² 15806	
Radial Bearings BABBITTED Backing STEEL		Loading, kPa TBA	
Thrust Bearing(s) BABBITTED Backing STEEL		Rating, Kpa 1700	
SHAFT END DETAIL		Int. Thrust Load, N (+)(-) 0	
☒ Gear Shaft End for Coupling Integral Flange (2.5.4.2)		Ext. Thrust Load, N (+)(-) TBA	
☐ Pinion Shaft End for Coupling Integral Flange (2.5.4.2)		Power loss each, kw 1,7	
☐ Other		Oil flow each, l/min 13,20	
		NOTES: The gearbox is delivered with a retaining nut for tapered end shaft	

SPECIAL PURPOSE GEAR UNITS **API 613 FIFTH EDITION** **DATA SHEET** **SI UNITS**

Job No. 01864H1 Item No. VV-C-5283001
P.O. No. 440591464 Date 20-janv-23
Doc No. YOH000090231 Rev 6 Date 24/11/23 By LFK

● LUBRICATION REQUIREMENTS Oil System Furnished By (2.8.3): Other BHNP ☐ Gear Vendor ☐ Oil Visc.: 29 cP @ 40°C 5 cP @ 100°C (2.8.6) ISO Grade VG32 Load Stage (2.2.6) 5				○ ADDITIONAL REQUIREMENTS ○ VIBRATION DETECTORS RADIAL (3.4.2.1) (2.7.4.5) (2.7.4.6) Manufacturer BENTLY NEVADA ☐ Total No. 8 X-Y Probes Pinion bearing: ☒ Coupling Enc ☒ Blind End X-Y Probes Gear bearings ☒ Coupling Enc ☒ Blind End Other ☐ AXIAL (3.4.2.1) (2.7.4.5) (2.7.4.6) Manufacturer BENTLY NEVADA ☒ Total No. 2 Dual Probes at each thrust bearing ☐ Single Probe any shaft without thrust bearing ☐ Other ☐ ONE EVENT PER REVOLUTION PROBE (2.7.4.5) (2.7.4.6) Manufacturer BENTLY NEVADA ☒ Total No. 2 One on Input Shaft ☒ One on Output Shaft ☐ Other ☐ ACCELEROMETER (3.4.2.1) (2.7.4.5) (2.7.4.6) Manufacturer BENTLY NEVADA ☐ No. Required 2 Pinion Coupling End ☒ Gear Coupling End ☒ Other ☐																											
■ MESH Mesh and windage power loss 7,2 kw Oil Flow, Mesh 0,70 m³/h 11,7 l/m				● TEMPERATURE DETECTORS ☐ Dial Type Thermometers (3.4.2.7) Type Brg. Temp. Sensors (3.4.2.7) (2.7.1.3) PT100 RTD ☐ Thermocouple / ☐ Simplex ☒ Duplex Calibration HS/LS Bearings no. sensors each 2 / 2 Thrust - number of sensing elements each face 2+2																											
■ LUBRICATION REQUIREMENTS Min. Startup Oil Temperature 21 °C Normal Oil Inlet temperature 50 °C Maximum Oil Inlet Temperature 60 °C Unit Oil Flow (Total) 5,04 m³/h 84 l/m Unit Oil Pressure 150 kPa Oil Visc.: 29 cP @ 40°C 5 cP @ 100°C (2.8.6) ISO Grade VG32 Load Stage (2.2.6) 5				● OTHER VIBRATION AND TEMPERATURE (3.4.2.5)(3.4.2.6) Other Gear Vendor Oscillator Demodulators Supplied By ☐ ☒ Vibration Monitor Supplied By ☒ ☐ Vibration Shutdown Delay Time Seconds Temperature Monitor Supplied By ☒ ☐ Oscillator Demodulator J-box By ☒ ☐ Temp. Sensor Termination J-box By ☒ ☐ J-box Type: Mount																											
■ PIPING CONNECTIONS <table border="1"> <thead> <tr> <th>Service</th> <th>No.</th> <th>Size</th> <th>Type</th> </tr> </thead> <tbody> <tr> <td>Lube Oil Inlet</td> <td>1</td> <td>1"1/2</td> <td>ANSI #150 LBS</td> </tr> <tr> <td>Lube Oil Outlet</td> <td>1</td> <td>6"</td> <td>ANSI #150 LBS</td> </tr> <tr> <td>Casing Drain*</td> <td>N/A</td> <td>N/A</td> <td>N/A</td> </tr> <tr> <td>Vent</td> <td>1</td> <td>2"</td> <td>ANSI #150 LBS</td> </tr> <tr> <td>Casing Purge</td> <td>2</td> <td>1"1/2</td> <td>NPT</td> </tr> </tbody> </table> *Casing drain is 'dead bottom' housing drain, not lube oil outlet				Service	No.	Size	Type	Lube Oil Inlet	1	1"1/2	ANSI #150 LBS	Lube Oil Outlet	1	6"	ANSI #150 LBS	Casing Drain*	N/A	N/A	N/A	Vent	1	2"	ANSI #150 LBS	Casing Purge	2	1"1/2	NPT	○ MISCELLANEOUS Undamped Critical Analysis Report (2.6.2.1): w/ Dampd Rotr Respns Analys Rprt (2.6.2.4.3) (2.6.2.6) Torsional Analysis By (2.6.5.2) ☐ Gear Vendor ☒ Other Spare Set Of Gear Rotors (4.3.2.4) Gear Case Furnished With Inlet Purge Connection (2.4.3) Orientation Of Oil Inlet & Drain Conns. (2.4.6) Filter Breather Location (2.3.1.12) TOP CASING Torsional Device Provisions (2.5.4.5) Rotor Vertical Storage Provisions (2.5.4.8) Rotor Vertical Storage Fixture(s) (3.6.2) Vendor Service Rep. on Site (2.1.13)			
Service	No.	Size	Type																												
Lube Oil Inlet	1	1"1/2	ANSI #150 LBS																												
Lube Oil Outlet	1	6"	ANSI #150 LBS																												
Casing Drain*	N/A	N/A	N/A																												
Vent	1	2"	ANSI #150 LBS																												
Casing Purge	2	1"1/2	NPT																												
■ MOUNTING PLATES ☐ Gear Furnished w/ (3.3.1.1) ☐ Baseplate ☐ Soleplate ☐ Subplate(s) (3.3.2.10) Mounting Plate(s) Furnished By (3.3.1.1) BH NP ☐ Baseplate Leveling (3.3.2.3) ☐ Baseplate With Leveling Pads (3.3.2.3) ☐ Baseplate Suitable For Column Mounting (3.3.2.4) ☐ Vendor Review of Purchaser's Foundation Dwgs. (2.1.13) ☐ Grout Type (3.3.1.2.4):				● CONTRACT DATA ☐ Test Data Prior To Shipment ☐ Progress Reports (5.3.3) ☐ Vendor Signoff of Inspector Checklist (4.1.1.3) ☒ Information Retained for 20 years (4.2.1.1) (4.2.6.1) ☐ Technical Manual (5.3.5.4) ☒ PAINTING (4.4.3.1) ☐ I-ET-5270.00-52313-97A-BI4-101 ☐ Painting Housing Interior not Allowed (2.3.1.14)																											
○ SHIPMENT Steel Rotor Storage Container (4.4.5) ☐ Shaft Covers (4.4.4) Contract Unit Spares Export Boxing ☒ ☐ Domestic Boxing ☐ ☐ Outdoor Storage Over 12 Mos. ☒ 3 yr Indoor ☒ (4.4.3.9)																															

SPECIAL PURPOSE GEAR UNITS **API 613 FIFTH EDITION** **DATA SHEET** **SI UNITS**

Job No. 01864H1 Item No. VV-C-5283001
P.O. No. 440591464 Date 20-janv-23
Doc No. YOH000090231 Rev 6 Date 11/12/2023 By LFK

● INSPECTIONS AND TESTS (4.1)					NOTES:
● Advance Notice of Witness Testing Required (4.1.3)					
Number Calendar Days 45					
	Req'd	Wit-ness	Ob-serve	Test Log	
6 Shop Inspection (4.1.1)	●	●	○		2. Gear casing material S275 J0 is equivalent (but not exactly) to ASTM 1011 / 1011 M SS Grade 40.
7 Cleanliness Inspection (4.2.3.1)	●	●	○		
8 Hardness Verification (4.2.3.3)	●	○	○	●	
9 Dismantle-Reassembly (4.3.2.3.1)	●	●	○	○	
10 Contact Check (2.5.2.2)	●	●	○	●	
11 Contact Check Tape Lift (2.5.2.2)	○	●	○	●	
12 Gear Accuracy Check (2.5.2.1)	●	●	○	●	
13 Double Helical Axial Stability (2.5.2.3)	○	●	○	●	
14 Special testing integral forged gears (2.5.3.3)					
15 (testing per mutual agreement)	○	○	○	○	
16 Residual Unbalance Chk. (2.6.6.2)	●	○	○	●	
17 Mechanical Run Test (Main) (4.3.2)	●	●	○	●	
18 Mech. Run Test (Spare) (4.3.2.4)	●	○	○	○	
19 Add'l. Mechanical Tests (4.3.2.2.16)	○	○	○	○	
20 Part Or Full Load And Full Speed					
21 Test (4.3.3.1)	○	○	○	○	
22 Full Torque, Reduced Speed (4.3.3.2)	○	○	○	○	
23 Full Torque Static Test (4.3.3.3)	○	○	○	○	
24 Back-To-Back Locked Torque					
25 Test (4.3.3.4)	○	○	○	○	
26 Sound Level Test (4.3.3.5)	●	○	○	●	
27 Additional Gear Tooth Test (4.2.2.8)	○	○	○	○	
28 Use Shop Lube System	●	○	○		
29 Use Job Lube System	○	○	○		
30 Use Shop Vibration Probes, Etc.	○	○	○		
31 Use Job Vibration Probes, Etc.	●	○	○		
32 Final Assembly, Maintenance &					
33 Running Clearance (4.2.1.1.e)	●	○	○	●	
34 Oil System Cleanliness (4.2.3.2)	●	○	○		
35 Oil System-Casing Joint					
36 Tightness (4.3.2.1.4)	●	○	○		
37 Warning And Protection					
38 Devices (4.3.2.1.5)	●	○	○		
39 Seismic Vibration Data (4.3.2.2.3)	●	○	○	○	
40 Vibration, Phase Plots (4.3.2.2.12)	●	○	○	○	
41 Oil Inlet Range Test (4.3.2.2.13)	●	○	○	○	
42 Tape Recorded Vibration Data	○	○	○	○	
43 (4.3.2.2.14)(4.3.2.2.15)	○			Purchaser copy	
NON-DESTRUCTIVE TESTING (4.2.1.2) (4.2.2.1)					
45	Surface	Sub-Surface		Log	
46	○ Casing	○		○	
47	○ Rot. Elemts.	○		○	
48	○ Bearings	○		○	
49	○ Other :	○		○	
50	(Specify)				